

A Robust Superpolynomial Better-Response Trajectory in a Smoothed Network Congestion Game

Corrected partial progress on unrestricted smoothed convergence

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Status. Relative to the open questions stated by Giannakopoulos, Grosz, and Melissourgos, this note gives *partial progress, not a solution*. It does not settle the smoothed complexity of computing exact pure Nash equilibria outside their restricted classes, and it gives no lower bound for their canonical shortest-path best-response dynamics. It does prove a complete negative result for a nearby, overbroad formulation: a polynomial smoothed iteration bound cannot hold simultaneously for succinct network strategy spaces and *every* polynomial-time better-response pivoting rule.

Abstract

An earlier draft proposed a one-player, constant-latency network example with exponentially many paths. The total-order observation behind that example is valid, but the claimed connection to the source paper is not: the source paper treats explicitly represented congestion games under arbitrary better responses and treats succinct network games under canonical shortest-path best responses. Moreover, the earlier draft did not exhibit a conventional polynomial-time pivot rule that realizes its perturbation-dependent ordering.

We repair the latter defect and state the resulting contribution at the correct level. For every $k \geq 1$, we construct a polynomial-latency network congestion game with $k + 1$ players, $5k + 1$ vertices, $6k$ edges, and $k^2 + 7k$ independently perturbed coefficients. Every coefficient has support in $[0, 1]$ and density at most 4. There is a fixed initial profile and a memoryless $O(k)$ -time better-response pivot rule whose trajectory has exactly $2^k - 1$ moves for *every* realization of the coefficients. The construction uses polynomial degrees at most $2k$ to create a robust superincreasing sequence of branch-cost gaps. Thus the expected trajectory length is superpolynomial in the succinct input size even for constant smoothing parameter.

This is a genuine barrier to extending pivot-rule-robust convergence guarantees from explicit strategy lists to arbitrary better responses over succinct path spaces. It is not a hardness result for equilibrium computation: on the same instances, shortest-path best-response dynamics reach the unique relevant best response in one move. Accordingly, the result is partial progress toward delineating the boundary of the source paper, not a resolution of its unresolved exact-equilibrium questions.

1 Scope and relation to the source paper

Giannakopoulos, Grosz, and Melissourgos (GGM) develop smoothed bounds for local search that are robust to the initial state and the pivoting rule, and apply them to congestion games under several latency representations [1]. Two different strategy representations must be kept separate.

For their “vanilla” congestion-game model, every player’s strategy set is listed explicitly. Single-player deviations can therefore be searched by scanning a neighborhood whose size is polynomial in that explicit description. Their arbitrary better-response theorem applies to games satisfying a bounded-change condition (B -restrained games).

For network congestion games, strategies are represented implicitly as source–sink paths and can be exponentially numerous. GGM therefore use a shortest-path oracle: a selected player moves to a minimum-cost path. They call these canonical dynamics *shortest-path dynamics* and prove smoothed bounds for (A, B) -compact network games. Their conclusion leaves the exact-equilibrium complexity outside the restricted general and network classes unresolved [1].

The secondary webpage that motivated the earlier draft ¹ is itself labeled “not verified as open,” and its proposed solution attempts are labeled unverified. Its broad wording should therefore not be substituted for the precise models in the source paper. The relevant distinctions are summarized below.

Setting	Strategy representation	Dynamics	Status
GGM, general games	Explicit strategy lists	Arbitrary better responses	Smoothed bounds under bounded restraint
GGM, network games	Implicit source–sink paths	Shortest-path best responses	Smoothed bounds under compactness
This note	Implicit source–sink paths	Arbitrary better responses	One efficient rule has a superpolynomial trajectory

Our theorem therefore has a deliberately limited interpretation. It rules out a universal-pivot extension in the third row. It neither refutes nor resolves either of the first two rows outside the parameter ranges analyzed by GGM.

2 Model and the claim proved

A directed network congestion game consists of a directed graph $G = (V, E)$ and a finite set of players. Player p has an origin o_p and destination d_p , and her strategies are the simple directed o_p – d_p paths. For a strategy profile $\mathbf{P}$, let $\ell_e(\mathbf{P})$ be the number of players whose paths contain edge e . If edge e has latency function κ_e , then player p ’s cost is

$$C_p(\mathbf{P}) = \sum_{e \in P_p} \kappa_e(\ell_e(\mathbf{P})).$$

A unilateral deviation is a *better response* if it strictly lowers the deviating player’s cost. A profile with no better response is a pure Nash equilibrium. Rosenthal’s exact potential changes by exactly the deviating player’s cost change, so every strict better-response path is finite [3].

A pivoting rule is an algorithm that, given the current realized instance and current profile, either certifies that the profile is a pure Nash equilibrium or returns an improving unilateral deviation. Following the standard PLS requirement used by GGM, the rule must run in time polynomial in the instance and profile description. A rule is *memoryless* if its output depends only on that input, not on a stored history of earlier moves.

¹Open Problems in Operations Research, item 28.

We use GGM's polynomial-latency smoothing model. Each edge has

$$\kappa_e(\ell) = \sum_{j=0}^{D_e} \alpha_{e,j} \ell^j, \quad \alpha_{e,j} \geq 0,$$

and all coefficients are independently sampled from distributions supported in $[0, 1]$, each having density at most a common parameter ϕ .

Theorem 2.1 (Main result). *For every integer $k \geq 1$, there is a smoothed polynomial-latency network congestion game $\mathcal{G}_k$ with the following properties.*

- (i) $\mathcal{G}_k$ has $k + 1$ players, $5k + 1$ vertices, $6k$ edges, and maximum latency degree $2k$.
- (ii) It has exactly $k^2 + 7k$ independently perturbed coefficients, all supported in $[0, 1]$ with density at most $\phi = 4$.
- (iii) There is a fixed, perturbation-independent initial profile and a memoryless $O(k)$ -time better-response pivoting rule that performs exactly $2^k - 1$ moves before reaching a pure Nash equilibrium, for every realization of the coefficients.
- (iv) The longest better-response path from that initial profile also has exactly $2^k - 1$ moves.

Consequently, both the realized and expected iteration counts are superpolynomial in the succinct input size, despite constant ϕ .

The construction and proof occupy Sections 3–5.

3 The network construction

Fix $k \geq 1$. There is one strategic player, denoted 0, and k dummy players, denoted $1, \dots, k$. The graph consists of k serial stages. Stage i has two internally disjoint three-edge routes from v_{i-1} to v_i :

$$v_{i-1} \rightarrow a_i \rightarrow b_i \rightarrow v_i \quad \text{and} \quad v_{i-1} \rightarrow c_i \rightarrow z_i \rightarrow v_i.$$

Write

$$\begin{aligned} p_i &= (v_{i-1}, a_i), & e_i &= (a_i, b_i), & q_i &= (b_i, v_i), \\ r_i &= (v_{i-1}, c_i), & s_i &= (c_i, z_i), & t_i &= (z_i, v_i). \end{aligned}$$

Player 0 has origin v_0 and destination v_k . Dummy player i has origin a_i and destination b_i . Since e_i is the only directed a_i – b_i path, dummy i has the singleton strategy $\{e_i\}$.

Thus player 0 has exactly 2^k paths, one for each independent choice of an upper or lower route in every stage. Encode such a path by $x = (x_1, \dots, x_k) \in \{0, 1\}^k$, where $x_i = 0$ means that the upper route is used and $x_i = 1$ means that the lower route is used. The fixed initial profile has $x = 0^k$; every dummy uses its unique path.

The latencies are sampled as follows, independently across all coefficients.

- Every edge $f \notin \{e_1, \dots, e_k\}$ has a constant latency $\kappa_f(\ell) = \beta_f$, where $\beta_f \sim \text{Unif}[0, 1/4]$.
- Edge e_i has degree $D_i = 2i$ and latency

$$\kappa_{e_i}(\ell) = \sum_{j=0}^{2i} \alpha_{i,j} \ell^j, \quad \alpha_{i,j} \sim \text{Unif}[1/2, 1].$$

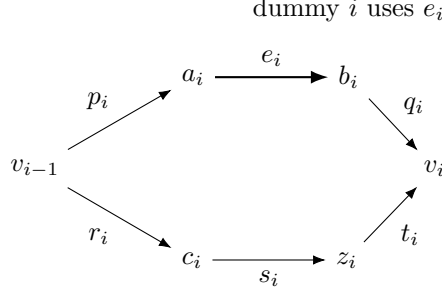


Figure 1: One stage of the construction. Player 0 chooses the upper or lower route; dummy i permanently uses the middle upper edge e_i .

The corresponding densities are 4 and 2, respectively, so the common density bound is $\phi = 4$. All coefficients lie in $[0, 1]$, and all latency functions are nonnegative and nondecreasing.

When player 0 takes the upper route in stage i , edge e_i has load 2; when player 0 takes the lower route, e_i has load 1 but player 0 does not pay for it. Define player 0's two stage costs

$$\begin{aligned} U_i &:= \beta_{p_i} + \kappa_{e_i}(2) + \beta_{q_i}, \\ L_i &:= \beta_{r_i} + \beta_{s_i} + \beta_{t_i}, \end{aligned}$$

and the saving from choosing the lower route,

$$w_i := U_i - L_i.$$

4 Robust superincreasing branch savings

The key point is that the ordering below holds for every realization in the stated supports. It is not a high-probability event.

Lemma 4.1 (Uniform bounds on one stage). *For every $i \in [k]$ and every realization of the coefficients,*

$$4^i - \frac{5}{4} \leq w_i \leq 2 \cdot 4^i - \frac{1}{2}.$$

In particular, $w_i > 0$.

Proof. Since $D_i = 2i$ and every $\alpha_{i,j} \in [1/2, 1]$,

$$\begin{aligned} \frac{1}{2} \sum_{j=0}^{2i} 2^j &\leq \kappa_{e_i}(2) \leq \sum_{j=0}^{2i} 2^j, \\ 4^i - \frac{1}{2} &\leq \kappa_{e_i}(2) \leq 2 \cdot 4^i - 1. \end{aligned}$$

The two other upper-route edges contribute between 0 and $1/2$, while the three lower-route edges contribute between 0 and $3/4$. Subtracting the latter cost from the former gives the stated interval. $\square$

Lemma 4.2 (Superincreasing separation). *For every realization and every $i \in [k]$,*

$$w_i > \sum_{j=1}^{i-1} w_j.$$

Proof. The claim is immediate for $i = 1$. For $i \geq 2$, Lemma 4.1 gives

$$\sum_{j=1}^{i-1} w_j \leq \sum_{j=1}^{i-1} \left(2 \cdot 4^j - \frac{1}{2} \right) < 2 \sum_{j=1}^{i-1} 4^j = \frac{2(4^i - 4)}{3} < \frac{2}{3} 4^i.$$

For every $i \geq 1$,

$$\frac{2}{3} 4^i < 4^i - \frac{5}{4},$$

because $4^i/3 > 5/4$. A final application of Lemma 4.1 proves the strict inequality. $\square$

Let

$$W(x) := \sum_{i=1}^k x_i w_i.$$

Because stages use disjoint edges, player 0's cost at path x is

$$C_0(x) = C_0(0^k) - W(x). \quad (1)$$

The next lemma identifies the complete cost ordering.

Lemma 4.3 (Binary order equals decreasing cost). *Define*

$$\text{val}(x) := \sum_{i=1}^k x_i 2^{i-1},$$

so stage 1 is the least significant bit. For all distinct $x, y \in \{0, 1\}^k$,

$$\text{val}(x) < \text{val}(y) \iff W(x) < W(y) \iff C_0(x) > C_0(y).$$

Proof. Assume $\text{val}(x) < \text{val}(y)$ and let h be the largest index at which x and y differ. Then $x_h = 0$ and $y_h = 1$. Hence, by Lemma 4.2,

$$W(y) - W(x) = w_h + \sum_{j < h} (y_j - x_j) w_j \geq w_h - \sum_{j < h} w_j > 0.$$

Interchanging x and y proves the converse. The cost equivalence follows from (1). $\square$

In particular, all 2^k path costs are distinct for every realization, not merely almost surely.

5 An efficient binary-counter pivot rule

Define a rule $\mathcal{R}_k$ on the profiles of $\mathcal{G}_k$ as follows. If player 0 currently uses $x \neq 1^k$, replace her path by the path encoding the integer $\text{val}(x) + 1$. If $x = 1^k$, halt. Equivalently, find the least index h with $x_h = 0$, set x_h to 1, reset $x_1, \dots, x_{h-1}$ to 0, and leave all more significant bits unchanged.

Lemma 5.1 (Validity and complexity of the rule). *The rule $\mathcal{R}_k$ is memoryless, runs in $O(k)$ time, and returns a strict better response at every nonterminal profile of $\mathcal{G}_k$.*

Proof. The current path determines its k branch bits. Scanning the bits and writing the successor path takes $O(k)$ time and no persistent state. If the successor is y , then $\text{val}(y) = \text{val}(x) + 1$, so Lemma 4.3 gives $C_0(y) < C_0(x)$. Only player 0 changes strategy, hence this is a strict unilateral better response. $\square$

Lemma 5.2 (The terminal profile is an equilibrium). *The profile with $x = 1^k$ is a pure Nash equilibrium. It is player 0's unique best response to the dummy players.*

Proof. Every w_i is positive by Lemma 4.1; therefore the lower route is strictly cheaper than the upper route in every stage, independently of all other stage choices. Thus 1^k uniquely minimizes player 0's path cost. Every dummy has a singleton strategy set and cannot deviate. $\square$

Proof of Theorem 2.1. The graph has the $k+1$ backbone vertices $v_0, \dots, v_k$ and four internal vertices per stage, for $5k+1$ vertices in total. It has six edges per stage. There are k dummies and one strategic player. The maximum degree is $D_k = 2k$.

The five noncentral edges in every stage each contribute one degree-zero coefficient. The central edge e_i contributes $2i+1$ coefficients. Hence the number of random coefficients is

$$5k + \sum_{i=1}^k (2i+1) = 5k + k(k+1) + k = k^2 + 7k.$$

The density bound $\phi = 4$ was verified in Section 3.

Starting from 0^k , Lemma 5.1 shows that $\mathcal{R}_k$ visits the paths with binary values

$$0, 1, 2, \dots, 2^k - 1$$

in this order, using one strict better response per increment. It therefore makes exactly $2^k - 1$ moves and then reaches the equilibrium of Lemma 5.2. This statement holds for every coefficient realization, so its expectation is the same.

Lemma 4.3 also shows that the better-response graph on player 0's paths is the complete acyclic graph oriented by binary value: from x there is a better-response edge to every y with $\text{val}(y) > \text{val}(x)$. No strictly improving trajectory can visit a profile twice, while $\mathcal{R}_k$ visits all 2^k profiles. Therefore the longest path from 0^k has exactly $2^k - 1$ moves.

Finally, the graph and player data have size polynomial in k , and the coefficient list has $\Theta(k^2)$ entries. Under a conventional binary encoding the total bit length is $O(k^2 \log k)$; under a word encoding it is $\Theta(k^2)$. In either case the input size is polynomial in k , whereas $2^k - 1$ dominates every polynomial in that size. Thus the iteration count is superpolynomial. $\square$

Remark 5.3 (A pivoting rule for the whole problem class). If one insists that a pivoting rule be a single algorithm defined on all network congestion games rather than on the displayed family, extend $\mathcal{R}_k$ as follows. Recognize the graph-and-player pattern above. If a binary successor exists, compute its realized cost and take it whenever it is strictly improving; otherwise, scan the players and use the standard polynomial-time shortest-path oracle to find a best response, or certify that none exists. Recognition, cost evaluation, and the shortest-path computations are polynomial. On every realization of $\mathcal{G}_k$, the binary successor always passes the improvement check, so this single rule for the full class has exactly the trajectory proved above.

6 Why this is not a solution of the GGM open problem

Theorem 2.1 is deliberately a theorem about *arbitrary better responses in a succinct network representation*. It does not show that exact pure Nash equilibria are hard to compute, nor that GGM's shortest-path dynamics are slow. Indeed, it demonstrates the opposite on this family.

Proposition 6.1 (Shortest-path dynamics terminate immediately). *From the fixed initial profile of $\mathcal{G}_k$, a shortest-path best response by player 0 moves directly to 1^k . Hence shortest-path dynamics require one improving move.*

Proof. By Lemma 5.2, the lower route is strictly cheaper in every stage. The unique minimum-cost v_0-v_k path therefore takes every lower route, namely 1^k . $\square$

The family is $(1, 3k)$ -compact in GGM’s terminology: every player has a unique possible best-response path, and player 0’s has length $3k$. Its polynomial latency degree is $2k$. Thus it lies outside the constant-parameter regime yielding the clean polynomial corollaries in GGM, but it is not a counterexample to their theorem. Likewise, viewing all paths as an explicit general-game strategy set would require listing 2^k strategies, in which case a 2^k -step path would no longer be superpolynomial in the explicit description.

The precise implication of our result is the following.

Corollary 6.2 (Barrier to a universal-pivot extension). *There is no polynomial p such that, for every succinctly represented polynomial- latency network congestion game and every polynomial-time better-response pivoting rule, the expected number of moves from every initial profile is at most $p(N, \phi)$, where N is the succinct input size.*

Proof. Apply such a bound to the rule and fixed initial profiles of Theorem 2.1. Here $\phi = 4$, N is polynomial in k , and the expected number of moves is $2^k - 1$, a contradiction. $\square$

This corollary is a complete negative answer to the hybrid formulation obtained by combining a succinct network strategy representation with a guarantee required to hold for every better-response pivot. Relative to GGM’s actual unresolved questions, it is only partial progress: it identifies one extension that is impossible and explains why the network model must control the move oracle, the represented strategy set, or both.

7 Audit of the earlier draft

For completeness, we separate the sound observations in the earlier constant-latency draft from the invalid inferences.

7.1 What was correct

In any one-player game with M strategies and pairwise distinct realized costs, every other strategy is a unilateral deviation. The better-response graph is therefore the complete acyclic orientation induced by cost, and its longest path has exactly $M - 1$ edges. A serial-diamond network with k stages has 2^k source–sink paths, and independent atomless constant edge costs make all path costs distinct almost surely. Consequently, the *longest transition-graph path* in that hybrid one-player model has length $2^k - 1$ almost surely. The symmetry argument giving expected maximum continuation length $(2^k - 1)/2$ from a fixed branch-symmetric path is also correct.

7.2 What was not established

Three points prevented that draft from being a valid solution of the source problem.

- (1) **Model mismatch.** The draft combined exponentially many implicitly represented network paths with unrestricted moves to any cheaper path. GGM’s network theorem instead analyzes shortest-path best responses; their arbitrary better-response theorem uses explicit strategy lists.

- (2) **No standard efficient predecessor pivot.** The rule “move to the highest-cost path below the current path” requires solving a nontrivial ordered subset-sum predecessor problem. The proposed priority-queue enumeration was a history-dependent procedure with a potentially exponential stored frontier, not a memoryless polynomial-time pivot rule given only the current profile.
- (3) **Input-size overstatement.** Under ordinary bit encodings, an $O(k)$ - object graph has $O(k \log k)$ rather than necessarily $\Theta(k)$ bits. The old example is still superpolynomial in a succinct encoding, but claims of $2^{\Omega(N)}$ require an explicit encoding convention.

The construction in this note repairs items (2) and (3): the successor is a fixed binary increment computable in $O(k)$ time, its improvement is robust for every realization, and the size accounting is explicit. It does not pretend to repair item (1). Instead, it turns that mismatch into a precisely stated boundary result. The price of obtaining a robust efficient rule is that the new theorem uses the polynomial-latency model with growing degree; it does not simultaneously cover the general-latency and step-function models.

8 Conclusion and open directions

The corrected result is a strong but narrow negative statement. Constant-density smoothing does not prevent superpolynomial exact better-response trajectories when an implicit path space is paired with unrestricted better responses: a fixed memoryless polynomial-time rule can traverse all 2^k paths, and the ordering survives every allowed perturbation. Thus pivot-rule robustness does not transfer automatically from explicit strategy lists to succinct network games.

This does *not* resolve the substantive exact-equilibrium questions left by GGM. In particular, the following remain outside the present proof:

- smoothed convergence of shortest-path best-response dynamics on unrestricted network congestion games;
- exact-equilibrium complexity for explicit general congestion games beyond the restrained regime;
- analogous robust lower bounds with constant polynomial degree, or in the independently perturbed general-latency or step-function models.

An FPTAS is known for approximate equilibria under the same broad smoothing philosophy [2]; the issue here is exact convergence. The correct status of this note is therefore *partial progress*: it closes an overbroad universal-pivot interpretation and supplies a rigorous obstruction, while leaving the authors’ main unresolved exact questions open.

References

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